



OPTIMA

AUTOMATED STUDY FLOW

USER MANUAL

Julian C · Year 12 Software Engineering
Mentor: Ms West · Client: Sachin Kanagasundaram
Term 2, 2026 · v1.0 · 29 May 2026

User Manual

Everything you need to know to use Optima as a day-to-day study planner. Please read this in order if it's your first time, or skip to the section you need.

CONTENTS

1. What Optima is for
2. Your first run
3. Creating an account & signing in
4. Setting up your subjects
5. Entering your classes & Fixed events
6. Adding assignments
7. Reading the dashboard
8. The weekly view
9. The monthly view
10. Preferences: theme, accessibility, notifications
11. Tips for daily use
12. Keyboard shortcuts
13. Adding a desktop shortcut
14. FAQ

1. What Optima is for

Optima is a study planner catered towards senior school students on a 14-day rotating timetable. It takes three main things from the user:

1. Your **classes** (where they fall in the 14-day timetable)
2. Your **assignments** (title, subject, due date, weighting, estimated hours).
3. How far you are through each one (a progress slider, updated as you work).

It uses that to give you three things back as well:

- + A **priority list** — what to work on next, with the most urgent being at the top.
- + A **generated weekly schedule** — specific study blocks dropped into the gaps in your timetable, sized 30 to 120 minutes with breaks between them.
- + A **cushion gauge** — how much spare study time you have over the next fortnight, expressed as a percentage of your remaining workload.

Optima runs offline on your own computer. None of the data leaves your machine.

2. Your first run

If you haven't installed Optima yet, follow the *Installation Guide* first. Once installed, please launch it from your system's terminal:

```
python3 run.py
```

A window titled **Optima — Automated Study Flow** should open. The splash screen will appear, where you will then be prompted to sign in or create a new account.

3. Creating an account & signing in

Sign up

1. On the splash or login screen, select **Create an account**.
2. Type a **username**. Allowed characters include letters, numbers, and any of the following: _ - @. Length: 3-32 characters.
3. Type a **password**. This must be at least 6 characters (and at most 128) and no required mix of symbols, so a memorable passphrase would be fine.
4. Re-type the password in the confirm field.
5. Click **Sign up**. You're now logged in and on the dashboard.

There is no email field in this process. By design, this also means there is no password reset, so make sure not to forget your password.

Sign in

1. On the splash screen, select **Sign in**.
2. Type your **username and password** in their respective fields.
3. Click **Sign in**.
4. If something's wrong, you'll see an error message, *"Invalid username or password"*. For security, it is not revealed which one was the problem.

Sign out

An **exit icon** is at the bottom left of the screen, at the bottom of the sidebar. You will be returned to the login screen and signed out.

Multiple accounts on one computer

Optima supports as many accounts as you would like to use on the same machine. Each one is stored in its own JSON file under `data/users/`, and they cannot read each other. This would be useful if you are using a shared computer or if you want a real and test account.

4. Setting up your subjects

Optima sets five default subjects the first time you log in: English, mathematics, software engineering, physics, and modern history. To change them:

1. Click **Preferences** in the sidebar.
2. Scroll to the **Subjects** card.
3. To delete a subject, click on the **x** on its chip.
4. To add a subject, type a name, optionally pick a colour (colours every assignment and study block for that subject), and click **+ Add Subject**.

TIP – PICKING DISTINCTIVE COLOURS

The weekly view shows class periods and study blocks colour-coded by subject. If two subjects share a similar colour, the grid becomes hard to read at a single glance. Although Optima won't stop you from doing it, you'll find it easier once you pick visually distinct HEX codes. Although Optima doesn't have a visual colour picker, you can reference online ones, like [Google's Colour Picker](#).

5. Entering your classes & fixed events

Classes, extracurricular activities, and any appointments you may have will all be added straight from the **Weekly View** rather than from the Settings page. The editor was redesigned in v1.1 because the old form in the preferences was slow, with nine inputs to add a single class, and no obvious way to fix a typo or mistake other than to delete and retype the subjects. Now you will be able to simply point, click and drag on the calendar.

Add a new event

1. Open the **Weekly** view from the sidebar.
2. Click an empty area in the grid on the day you want, and drag down to mark how long the event should be (snaps to 5-minute increments).
3. An *edit panel* opens with the times pre-filled. Fill in:
 - **Name:** Free text, e.g., "Maths"
 - **Type:** Subject (default setting), Extracurricular, Designated Study Block, or Other. A *study block* tells the auto-scheduler that it may put study sessions here. Classes, extracurriculars and other events are off-limits. (There was a separate 'Study period' tick-box, but it has been removed since it was redundant with the use of 'Designated Study Block' in the Type dropdown.)
 - **Subject:** Optional. Pick from the subjects you set up in step 4. Determines the colour.
 - **Use a preset:** If you've saved Period bell times in *Settings > Periods*, pick one here to populate the start/end of the event immediately. Saves the effort of typing.
 - **Repeats:** Default set to *every two weeks* (most schools use a 14-day rotation), but you can select from *Just this date*, *Every day*, *Every weekday (Mon-Fri)*, *Every week*, *Every month*, or *Every year*. "Every weekday" would be the right choice for a class or assembly that occurs every school day but not at weekends.

- **Start/End:** Times in the format your preference is set to, e.g., 9:30 pm in 12-hour mode or 21:30 in 24-hour mode. These are internally stored as decimal hours.
 - **First occurrence:** The calendar date the repeating rule should anchor against, pre-filled with the day you selected.
4. Click **Save**. The event, and every future version of it, appears on the grid immediately.

DESIGNATED STUDY BLOCKS

The automatic scheduler Optima runs will **prefer** to place generated study sessions inside events with the type "Study block". Sleep, classes, extracurricular commitments and other events will always be blocked off. There are a few exceptions, though:

- + **Escalation:** Homework due in ≤ 24 hours, or an exam or project due in ≤ 72 hours, may spill into any free time outside sleep and fixed events.
- + **Overflow:** After the main placement, anything still short of time is notified in the dashboard, stating how much extra time is needed. e.g., "Need an extra 30 mins for <task>".

Without these, the scheduler would refuse to help with a last-minute exam if you didn't have a designated time free, and as such, it does the right thing without going through impossible events like a bus commute under normal conditions.

BREAKS INSIDE LONG STUDY BLOCKS

A 4-hour designated study block won't get filled with a single 4-hour session cramming one task. Instead, Optima is designed to rotate these tasks (e.g., <task 1> for 2 hours, <task 2> for the remainder). However, if you hover over a day's window in the weekly view, the tooltip will list every break time and study time explicitly. The pattern of a 30-minute minimum session and 2-hour maximum, with a 15-minute break between consecutive sessions, will repeat for each study block.

Edit, move, or delete events.

- + **Edit:** click an event to open the editing panel. If the event repeats, you will be asked whether you want to apply your changes to *all occurrences* or *only this single one*.
- + **Move:** click and drag an event to a new time or day. It snaps automatically to 5-minute increments. For repeating events, drags only move the one instance (a per-day override is default, so the original rule is preserved).
- + **Multi-select:** hold Shift (or Cmd/Ctrl) while clicking to add additional events to the selection. Drag any selected event to move the whole group together. Press Delete to remove the selection.
- + **Delete a repeating event:** click it, then click **Delete**. You will be prompted to confirm: '*Only this one* (just skip this date)' or '*All occurrences* (forget the whole rule)'.

TIME FORMAT

Open **Preferences > Time format** to switch between 12-hour and 24-hour display modes. The setting only changes how times appear on screen, so nothing in your saved data is changed.

WHAT HAPPENS DURING LUNCH/RECESS/SPORT?

You can model these events with the *Other* type and no subject. By doing this, the scheduler won't place study blocks during them. This could be useful; the overflow of assignments could leak into your break time.

6. Adding assignments

An **assignment** in Optima is any piece of work with a due date and an estimated effort. Major assessments, weekly homework, depth studies, and exam revision – all of these are covered by creating an assignment.

1. From the dashboard, click **+ New task**.
2. Fill in:
 - **Name:** A short name, e.g. 'English Mod C', or 'Physics Stem 6-7'.
 - **Subject:** One of your subjects from step 4. When you pick a subject and the task type is *homework*, Optima will automatically fill the due date with the next occurrence of that subject's class, usually when that homework is due next lesson in practice. This can be edited later.
 - **Type:** The default is *Homework*; you may select *Exam/assessment* or *Project* instead. This determines the next two fields:
 - *Homework* and *projects* use a categorical **Importance** (Low/Medium/High; Default Medium)
 - The exam/assessment uses a **weighting** % from 0-100, which is how much of the course mark the task is worth. Useful when dealing with exam blocks.
 - **Due date and time:** When it has to be handed in. The date must be today or later; the time defaults to 09:00 if the subject is left blank. If a subject has been selected for the task, both fields will automatically prefill with the date and start time of your next class for that subject (editable).
 - **Estimated time:** Enter hours and minutes separately. e.g., 2h 30min, or 0h 45min for a quick revision session. The scheduler uses this to size study blocks. (Internally kept as a float).
 - **Show the stretch line on the monthly view:** by default. Draws a coloured bar across the monthly calendar from today's cell to the due-date cell in your subject's selected colour, so the days left are visible at a glance. Turn off for a minute tasks that you don't want cluttering the calendar.
3. Click **Create**.

HOW IMPORTANCE/WEIGHTING MATTERS

Internally, there is one priority formula: $\text{score} = (\text{scoring_weight} * 10) / \max(\text{hours_until_due} / 24, 1)$. For exams, `scoring_weight` is the entered %. For homework/projects, this comes from the importance map: Low = 10, Medium = 25, High = 50. Any task within **24 hours** of its due time jumps to a *critical* status; anything past the due time becomes *overdue* (red badge, sorted to the top, and the scheduler stops planning future blocks for it).

Tracking progress

On the dashboard, each task has a progress slider. Drag it as you complete the work. The slider saves the moment you let go. The cushion gauge updates the next time you load the dashboard.

Editing or deleting a task

Click the task title to open its edit page. You can change any field or click **Delete task** at the bottom. A confirmation prompt will appear before you delete.

7. Reading the dashboard

The dashboard is the home screen and the place you will likely spend most of your time. It has four main panels:

KPI strip (at the top)

KPI	What it means
Open tasks	Total number of unfinished assignments.
Due this week	How many of those are due within 7 days?
Hours scheduled	Sum of study-block lengths across the next 14 days.
XP	Lifetime experience points. Increments when tasks are marked as completed (100%). Purely cosmetic currently.

Priority tasks (left)

The list of your open assignments, sorted by priority. The top one is what Optima recommends you work on next. Each card will show:

- + A coloured strip on the left in the subject colour.
- + The title, due date, and weighting.
- + A critical badge if the task is due within 3 days or fewer.
- + A progress slider.

Today's schedule (bottom)

The list of generated study blocks for today, in time order. If today is empty and there are no scheduled blocks, you'll see a 'Nothing scheduled' message. It doesn't necessarily mean that there is nothing to do, but it means that today's free slots weren't large enough for the priority list, or the priority list is empty.

8. The weekly view

Click the **Blank Calendar Icon** in the sidebar.

You will see a 7-day grid for the current week, with a column per day, and it is divided hourly on the left. Each cell is filled with the things that happen within that hour:

- + **Solid-coloured blocks:** Classes, extracurriculars, and any appointments that you have added.
 - + **Dashed-coloured blocks:** Generated study blocks for your assignments.
 - + **🔄 Icon:** in the corner of a block means that it repeats. Click it once to edit just that instance, or use the panel's *Repeat* dropdown to change the repeating rule.
 - + **Today's column** is highlighted in pale purple, so it is easily findable at a glance.
-

9. The monthly view

Click the **Dotted Calendar Icon** in the sidebar.

A standard 6-row by 7-column calendar grid for the current month will appear. Each cell shows up to four due-date dots, colour-coded by subject, with one per assignment due that day. Hover over a dot to see the assignment title.

Above the dots, every active assignment draws a horizontal **stretch line** in its subject colour, running from today's cell to the assignment's due date cell. The bar allows you to visualise how much runway you have, which is answerable at a glance. (e.g., a long red maths-coloured bar tells you that the Maths exam is far out; a short yellow bar could tell you that English is due tomorrow.)

Each calendar row stacks up to 6 stretch bars vertically, and anything beyond that becomes a discreet *+ n more* badge on the affected cells. You can toggle the stretch line visibility for individual tasks on the task form, which is useful for low-importance assignments you don't want to crowd the view with.

Use the navigation arrows at the top right of the screen to navigate back and forward between months. The current day is outlined in purple.

The monthly view is the right tool to use for medium-term planning, spotting weeks where three assignments fall together, or noticing that your subject mix is unbalanced.

10. Preferences: theme, accessibility, notifications

Theme

Light is the default and recommended choice during the day. **Dark** reduces eye strain at night and is friendlier on OLED screens.

Time format

Toggle between **24-hour** (e.g., 14:30) and **12-hour** (e.g., 2:30 pm) display. Affects every clock label across the app, in the weekly view, the dashboard, and every form. Optima internally stores these as decimal hours, so switching back and forth doesn't affect the precision.

Term start

The calendar date your school's Week A, Day 0, falls on (if applicable). Fortnightly events will anchor against this so that the rotation lines up correctly. You can move this if your school adjusts its calendar, but existing events will keep their stored dates.

Sleep window

Two clock times: when you go to sleep, and when you wake up. The window is allowed to wrap past midnight (e.g., 22:30 to 06:30). Optima will never place automatically generated study sessions inside it, and the weekly view will draw a hatched grey overlay over the corresponding rows on every day so that the off-limits are visible at a glance. Accepts values like 22:30 or 22.5.

Periods (preset times)

If your school runs the same period times every day (e.g., Period 1 8:50 - 10:30, Period 2 11:00 - 12:40, Period 3 1:10 - 2:50), you can save these once here. Now, when you add an event on the weekly view, the **Use a preset** dropdown lets you pick a *period*, and the start/end fields fill themselves. Click the 'x' next to a period to remove the preset.

High contrast

Switches to a black-on-white palette with a single bold accent colour. This is designed for situations where the screen is washed out by glare (sitting in the sun, etc.).

Zoom

A percentage from 75% to 200% (default 100%) is set with the slider or Ctrl/Cmd +, -, and 0; the whole interface scales proportionally, so layouts will still hold together cohesively at higher zoom levels.

Notifications

Toggles inline banners on the dashboard when a task is in the critical window. This doesn't post to the operating system. This was deliberately avoided, as they would be disruptive, and the planning documentation outlined subtle reminders.

Autosave

Persists preference changes as you make them instead of saving them manually. On by default. Turn off if you want to experiment with settings without committing these changes.

11. Tips for daily use

Morning routines

1. Open Optima, glance at the cushion gauge (green, amber, red).
2. Read the top three items on the priority list.
3. Look at today's schedule to see which slots are pre-filled, checking your availability for the day.
4. Total time: around 30 seconds. You will now be ready for the day.

End-of-day updates

1. For each task you worked on today, drag the progress slider to the new estimated percentage.
2. If anything has changed (teacher modified due date or new assignments), update and add them now.
3. That's it. Tomorrow morning, your priority list and cushion gauge will capture what you have completed.

What to do when the cushion is red

The gauge is telling you that there isn't enough time for you to finish everything to the standard you'd want. Your options:

- + **Defer or drop** a low-weighting task. A piece of homework can be dropped in favour of a highly weighted task.
- + **Ask for an extension** on a major piece of work or project. Better to ask early than to hand in incomplete work or nothing at all.
- + **Lengthen your awake window** if you're chronically under, but reflect and be honest about whether late nights actually help you. Maintaining a good sleep schedule is important for your performance throughout the day.

Trust the priority, with one caveat

The formula (weighting * 10) / days remaining matches most students' intuition. The one place it falls short is when two tasks are both within the critical window. They will both show as a priority

(999), and the tiebreaker is their due date. If one of those is a 30% HSC task and the other a piece of homework, use your judgement.

12. Keyboard shortcuts

Keys	Action
Tab / Shift+Tab	Move between form fields and buttons.
Enter	Submit the currently focused form.
esc	Close a confirmation prompt or deselect events in the weekly view.
Cmd/Ctrl + Click	Toggle one event in the selection.
Shift + Click	Add another event to the current selection in the weekly view.
Delete / Backspace	Deletes the selected events (confirmation prompt before removing repeating ones).
Cmd , / Ctrl ,	Jump to the preferences page.
Cmd W / Ctrl W	Close the window, but keep the session signed in.

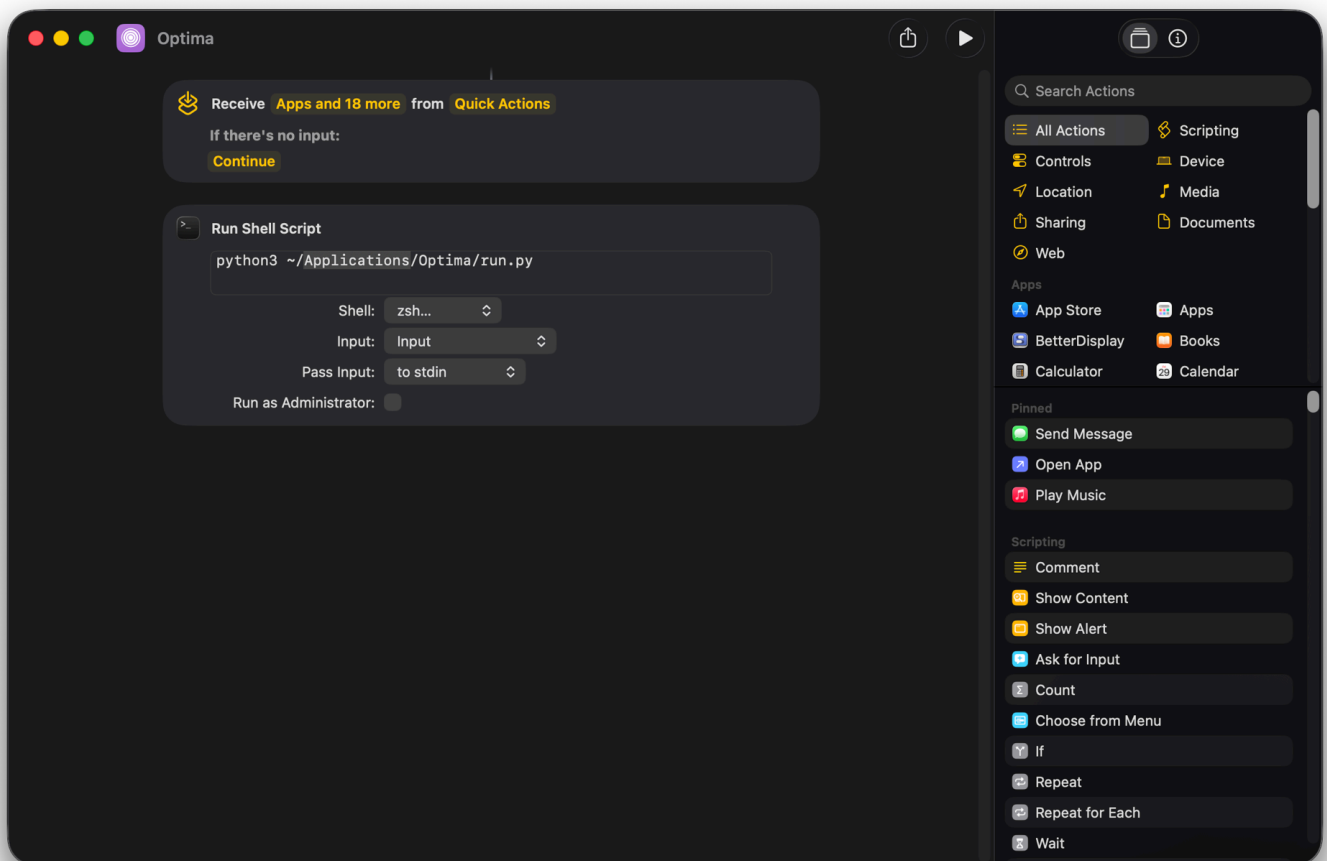
13. Adding a desktop shortcut

Typing ``python run.py`` into a terminal every time will be tedious. Here's how to launch Optima with a single click from your dock or desktop.

Step 1: Building the Shortcut

1. Open the **Shortcuts** app (Cmd + Space → type "Shortcuts").
2. Click **+** to create a new shortcut and name it Optima.
3. Search the actions list for **Run Shell Script** and add it.
4. Set **Shell** to `zsh` (should be by default), and enter this single line, adjusting the path if you've installed Optima somewhere other than `~/Applications/Optima`:

```
python3 ~/Applications/Optima/run.py
```



The Run Shell Script action in the Shortcuts app.

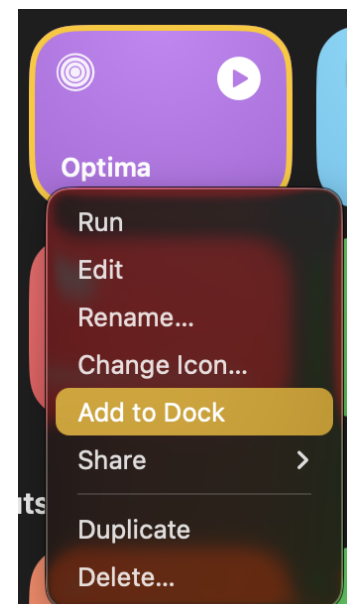
5. Press the ► button at the top-right to test it; the Optima window should open. If you've installed Optima's libraries into a virtual environment (as the installation guide recommends), use `~/Applications/Optima/.venv/bin/python` in place of `python3`.

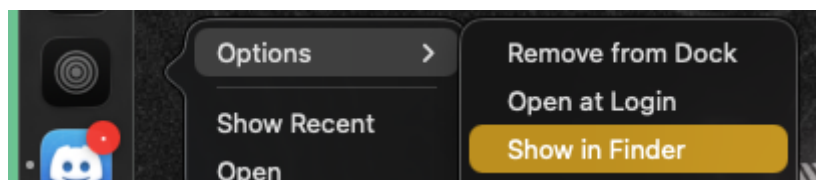
Step 2: Add it to the Dock

Right-click the shortcut (or open its ⋮ menu) and choose **Add to Dock**. An Optima button now lives in your dock.

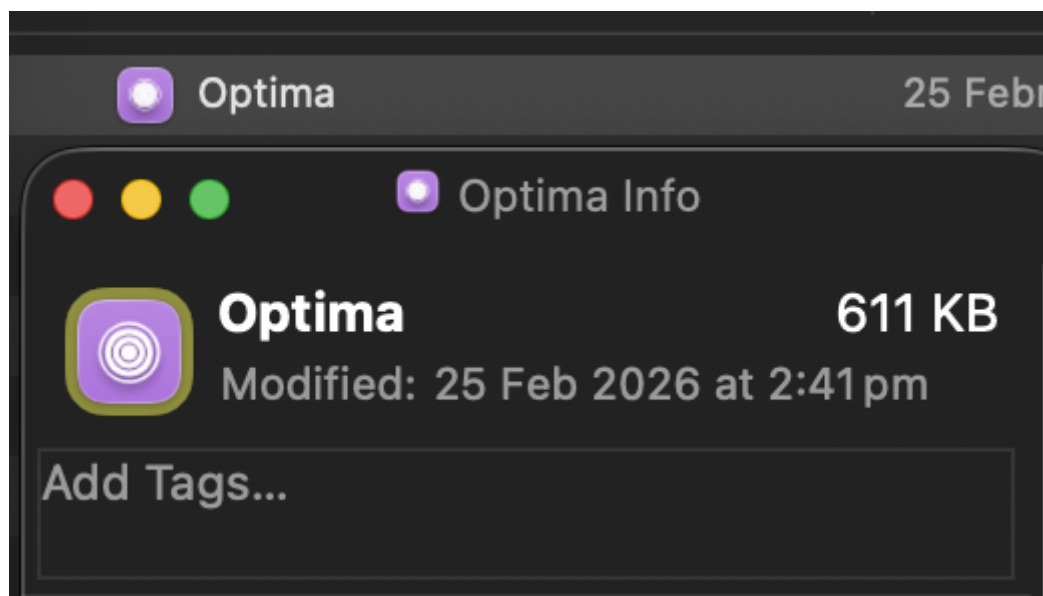
Step 3: Give it the Optima icon

1. **Open logo.png** with Preview (found in your Optima folder: `/Optima/img/logo.png`), then press `Cmd + A` then `Cmd + C` to copy the image.
2. Right-click the Optima icon in the Dock and choose **Options** → **Show in Finder**.





3. In Finder, **click the Optima file once** to select it, then press **Cmd + I** to open Get Info. Click the small icon in the top-left of the panel so it highlights, then press **Cmd + V** to paste.



4. Click the Optima button in the Dock once. Its icon refreshes to the Optima logo, and from now on, a single click launches the app.

14. FAQ

Is my data stored anywhere I can't see?

No. Everything is saved in `data/users/<your username>.json` inside the Optima folder. You can open the file in a text editor, and everything except for the password hash and salt is human-readable.

Can I move my data to another computer?

Yes. Copy your `data/users/<your username>.json` to the same path on the new computer. Sign in as normal.

Can two people share an account?

You can, but you probably shouldn't, as there may be different schedules and different progress on tasks. Just make a second account.

Why doesn't my assignment have a study block tomorrow?

One of three reasons:

- + It's not at the top of the priority list, and other assignments have filled the slots first.
- + Tomorrow's free slots are all shorter than 30 minutes (minimum session length).
- + You've already scheduled enough hours to finish it before its due date, and the scheduler doesn't add more.

If you're sure something's wrong, navigate back to the dashboard and back to the weekly view to regenerate the schedule. If the issue persists, please report it as an issue on the [GitHub page](#) to be worked on.

Can I export the schedule to Google Calendar?

Currently not in this version.

I forgot my password. What now?

Because Optima doesn't ask for an email at signup, there's no password reset. You can either create a new account with a different username and start fresh, or copy the user data stored in the json file (everything below and excluding username, password_hash and salt) into the new account's json file.

Does Optima know about public holidays/school holidays?

No. The scheduler assumes every day in the next 14 days is a potential study day with the same awake window. If you want to block out a holiday, add an 'other' constraint that covers the whole day on each affected day in the rotation.

Where do I report a bug?

You can report it as an issue on the [GitHub page](#) or email the developer directly. There is no in-app feedback link, deliberately, to avoid sending anything off the machine without your permission.